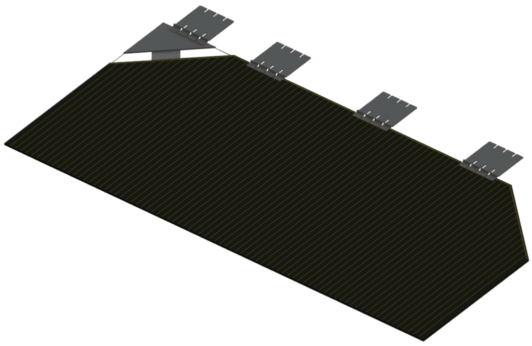
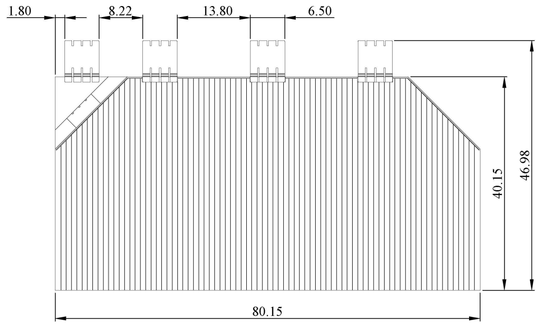
Triple-Junction GaAs Solar CICs

TJ3xSCA-4080



Size：40.15mm×80.15mm



Cell Aera：30.18cm²

Design and Mechanical Data

Substrate Material：	Ge
Base Material：	GaInP/GaAs/Ge
Thickness：	150μm±20μm
AR-coating：	TiO_x/Al_2O_3
Coverglass：	KFG120
Coverglass thickness：	120μm
CICs Size：	40.15mm × 80.15mm
CICs thickness：	270μm，300μm
Interconnector thickness：	30μm
Average Weight：	≤115mg/cm² Si
Bypass Protect：	Diode
Service orbit：	LEO, MEO, GEO

Extra option： 1. Coverglasses with different thickness can be customized.
2. ITO coverglass available.

Typical Performance Data

Electrical Parameters @ AM0（1353W/m2，T=25℃）		
Efficiency η_{bare} [%]	30	32
Open Circuit V_{oc} [V]	2.74	2.70
Short Circuit J_{sc} [mA/cm²]	17.4	18.7
Current @ Max. Power J_m [mA/cm²]	16.4	17.9
Voltage @ Max. Power V_m [V]	2.42	2.37

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Radiation Performance at 1Mev Electron Irradiation, EOL/BOL Ratios

Fluence (e/cm²)	1E14		5E14		1E15	
	30	32	30	32	30	32
V_{oc}/V_{oc0}	0.96	0.96	0.94	0.92	0.92	0.91
I_{sc}/I_{sc0}	0.99	0.99	0.96	0.97	0.95	0.94
P_{mp}/P_{mp0}	0.96	0.95	0.90	0.89	0.86	0.84

Bypass Diode

Forward Voltage（+2.5A）	＜1.0V
Reverse Current（-4.5V）	＜0.7mA

Interconnector

Type	Sliver	Kovar (sliver coated)
Pull test at 45° (one point)	＞1.6N	＞1.6N

Thermal Properties（CIC）

Absorption coefficient	≤0.89

Temperature Gradients

Type		30	32
Short Circuit J_{sc}	[μA/cm²/°C]	12.0	11.0
Open Circuit V_{oc}	[mV/°C]	-5.6	-6.0
Current at max. power J_m	[μA/cm²/°C]	9.0	10.0
Voltage at max. power V_m	[mV/°C]	-5.8	-6.3
Temperature Coefficient （20°C~80°C）			